

Tran Quoc Truong

Applied AI Engineer | Agentic QA & RAG Systems | Python Backend

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Summary

Applied AI Engineer building enterprise **Agentic QA, RAG**, and document-intelligence systems with production-oriented Python backends.

Strong in **Django/DRF, FastAPI, LangGraph, LangChain, LiteLLM, Redis/Celery, PostgreSQL, vector databases, MCP tool integration**, and QA automation.

Experience

FPT Software, Applied AI Engineer & QA Automation Engineer

Ho Chi Minh City, Vietnam

Dec 2024 – present

1 year 7 months

- Built **AI-assisted QA** workflows for document understanding, **RAG Q&A, test-case generation, duplicate bug detection, traceability**, and **change impact analysis**.
- Designed backend orchestration with **Django/DRF, FastAPI, Redis/Celery, PostgreSQL, async workers, service layers**, and clear **domain boundaries**.
- Developed document-processing flows for **OCR/preview** generation, enhanced **OCR** strategy, version commit, chunk generation, artifact storage, and vector indexing.
- Delivered API, Web, and Mobile automation solutions that increased test coverage by 40% and reduced manual testing effort by up to 70%.
- Led an omnichannel **Core Banking** automation PoC across Web, iOS, and Android, helping secure a ~20 man-month contract.
- Executed banking QA work across **Core Banking** and **Payment Hub** domains, including API testing, **SQL validation**, regression testing, and stress testing.

FPT Software Academy, Full-Stack Developer Intern

Ho Chi Minh City, Vietnam

Sept 2024 – Dec 2024

4 months

- Built Django-based web applications and deployed containerized services with Docker.
- Improved deployment reliability through production-oriented configuration and Cloud-flare integration.
- Reduced deployment cycle time by approximately 50% through cleaner containerized delivery workflows.

Projects

Omni-Agent - Enterprise AI Platform for Agentic QA****

Enterprise AI platform for turning documents, requirements, bugs, and test assets into searchable, traceable, and automatable QA workflows.

- Designed a multi-service architecture with **Django/DRF** as the **control plane**, **async workers** as the processing plane, and separate boundaries for **document processing**, retrieval, storage, graph analysis, and LLM orchestration.
- Modeled backend domains for Documents, Document Versions, Processing Artifacts, Chunks, and **Platform Jobs**, keeping API endpoints thin and workflow logic in **service layers**.
- Implemented versioned document workflows covering baseline upload, **OCR**/preview generation, enhanced **OCR** strategy, commit flow, chunk generation, artifact storage, and vector indexing.
- Integrated **RAG** infrastructure with **vector databases**, **LiteLLM** model routing, **MCP wrapper/tool integration**, **LangGraph/LangChain** workflows, and observability/evaluation patterns.
- Built **Agentic QA** capabilities for document-to-**test-case generation**, **RAG Q&A**, **duplicate bug detection**, requirement/test/defect **traceability**, and **change impact analysis**.
- Prototyped **React/Vite** frontend workflows with **AI-assisted development**, including project navigation, document workspace UI, preview tabs, version dropdowns, and processing status views.

APIT - Agent Programmatic Integration Testing

LLM-based system that analyzes API documentation and generates structured API test cases.

- Built the core **AI agent** for API document understanding, structured **test-case generation**, and automated output evaluation.
- Evaluated LoRA and QLoRA approaches for domain-specific structured test document generation.
- Fine-tuned Qwen-2.5-3B achieved macro-F1 0.655, outperforming Llama-3.2-3B by 41.5% in the evaluation setup.
- Implemented **FastAPI**-based model evaluation workflows and **LLM-as-a-Judge** experiments for generated output comparison.

Eximbank Core Banking

• Omnichannel Automation PoC

Automation PoC for validating end-to-end **Core Banking** workflows across Web, iOS, and Android.

- Built cross-platform automation flows with Katalon and Appium for omnichannel regression coverage.
- Validated critical business journeys consistently across Web, iOS, and Android under one automation strategy.
- Demonstrated stronger coverage and execution efficiency than competing vendors, helping secure a ~20 man-month contract.

Sacombank - Payment Hub Integration Testing

API, database, regression, and stress testing project for financial transaction integrations.

- Executed API testing, database testing, regression testing, and stress testing to validate transaction integrity and endpoint reliability.
- Authored more than 1,500 lines of **SQL validation** queries to verify post-transaction data consistency across payment integration flows.
- Reduced manual QA effort by 70% through reusable validation scripts, regression scenarios, and repeatable API/database checks.
- Maintained full API endpoint coverage while validating financial operation correctness across integration flows.

Skills

Applied AI, Agents & RAG: LangGraph, LangChain, MCP wrapper, RAG, Hybrid RAG, LiteLLM, Langfuse, LLM Evaluation, LLM-as-a-Judge, LoRA/QLoRA

Backend & AI Systems: Python, Django/DRF, FastAPI, PostgreSQL, Redis/Celery, Docker, REST APIs, async workers, service layers, domain boundaries

QA Automation & Banking Domain: Agentic QA, API Testing, Web/Mobile Automation, Katalon, Appium, Postman, JMeter, SQL validation, Core Banking, Payment Hub

Full-stack AI Prototyping: React, Vite, Tailwind CSS, shadcn/ui, AI-assisted development, dashboard UI, document workspace UI

Education

FPT School of Business & Technology, Master of Software Engineering in AI	Ho Chi Minh City, Vietnam May 2026 – present
FPT University, Bachelor of Artificial Intelligence	Ho Chi Minh City, Vietnam 2021 - 2025

Languages

Vietnamese: Native

English: Working proficiency

Japanese: N5 / Basic